1 Fundamental Theorem of Linear Maps

Theorem 1.1 (Fundamental Theorem of Linear Maps)**.** *Let $T : V \to W$ be a linear map with V finite-dimensional. Then imT is finite-dimensional and*

$$\dim V = \dim \ker T + \dim imT. \tag{1.2}$$

Exercise 1.3. Complete the following proof of Theorem 1.1.

Proof. Let $\{u_1, \ldots, u_m\}$ be a basis of $\ker T$. Then $\dim \ker T = m$. By the Plus/Minus Lemma, the linearly independent set $\{u_1, \ldots, u_m\}$ can be extended to a basis $\{u_1, \ldots, u_m, v_1, \ldots, v_n\}$ of V. Then $\dim V = m + n$. To complete the proof, we need to show that $\mathrm{im}T$ is finite-dimensional and $\dim \mathrm{im}T = n$.

(a) Show that $\{Tv_1, \ldots, Tv_n\}$ spans $\mathrm{im}T$.

(b) Show that $\mathrm{span}\{Tv_1, \ldots, Tv_n\}$ is also linearly independent and therefore a basis of $\mathrm{im}T$.

(c) Explain why this completes the proof.

□

Corollary 1.4. *Let V and W be finite-dimensional vector spaces, and let $T : V \to W$ be a linear map.*

(a) If $\dim V > \dim W$, then T is not injective.

(b) If $\dim V < \dim W$, then T is not surjective.

Exercise 1.5. Use Theorem 1.1 to prove Corollary 1.4.

Exercise 1.6. Consider the linear map $T : F^4 \to F^3$ defined by

$$T(z_1, z_2, z_3, z_4) = (\sqrt{7}z_1 + \pi z_2 + z_4, 97z_1 + 3z_2 + 2z_3, z_2 + 6z_3 + 7z_4).$$

Prove that T is not injective.

2 Systems of Linear Equations, Revisited

In the first unit, we represented a system of linear equations

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n &= y_1 \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n &= y_2 \\ \vdots \qquad\quad \vdots \qquad\qquad\quad \vdots \quad \vdots & \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n &= y_m. \end{aligned} \tag{2.1}$$

as a matrix equation

$$Ax = y. \tag{2.2}$$

As we saw last time, we can interpret (2.2) as saying that $y \in M^{m\times 1}(F)$ is in the image of the linear map $T : M^{n\times 1}(F) \to M^{m\times 1}(F)$ which multiplies x by the coefficient matrix A: $T(x) = Ax$.

Exercise 2.3. Use Theorem 1.1 to prove the following.

(a) A homogeneous system of linear equations with more variables than equations has nonzero solutions.

(b) A system of linear equations with more equations than variables has no solution for some choice of constant terms.

Exercise 2.4. Without doing any calculations, determine whether the following system is consistent for all b_1, b_2, b_3.

$$\begin{aligned} 3x_1 + 2x_2 &= b_1 \\ -2x_1 + 4x_2 &= b_2 \\ x_1 + 5x_2 &= b_3. \end{aligned}$$

3 Exercises

1. Suppose $T \in \mathscr{L}(F^4, F^2)$ such that $\ker T = \{(x_1, x_2, x_3, x_4) \in F^4 : x_1 = 5x_2 \text{ and } x_3 = 7x_4\}$. Prove that T is surjective.

2. Prove that there does not exist $T \in \mathscr{L}(\mathbb{R}^5)$ such that $\text{im} T = \ker T$.

3. Suppose $T \in \mathscr{L}(V, W)$ is injective and $\{v_1, \ldots, v_n\}$ is a linearly independent subset of V. Prove that $\{Tv_1, \ldots, Tv_n\}$ is a linearly independent subset of W.

4. Show that if $\{v_1,\ldots,v_n\}$ spans V and $T \in \mathscr{L}(V,W)$, then $\{Tv_1,\ldots,Tv_n\}$ spans $\operatorname{im} T$.

5. Show that $W = \{T \in \mathscr{L}(\mathbb{R}^5,\mathbb{R}^4) : \dim\ker T > 2\}$ is not a subspace of $\mathscr{L}(\mathbb{R}^5,\mathbb{R}^4)$.